Table 4: Partition sum of methane (Present versus HITRAN 2004)

	Present work			HITRAN 2004 [7]	HITRAN − Present
Temp. /K	Value	Uncertainty	Equiv. ΔT	(parsum.dat)	Percent
100	116.4	< 0.1 %	< 0.1	116.415997	0.01 %
200	326.6	< 0.1 %	< 0.1	326.631038	0.00 %
300	602.8	< 0.1 %	< 0.1	602.783207	-0.01 %
400	954.7	< 0.1 %	< 0.1	954.176097	-0.05 %
500	1417.7	< 0.1 %	< 0.1	1415.863977	-0.13 %
600	2045.7	< 0.1 %	< 0.1	2040.847920	-0.25 %
700	2910.7	< 0.1 %	< 0.1	2899.589576	-0.40 %
800	4109.1	< 0.1 %	< 0.1	4085.949715	-0.58 %
900	5770.1	< 0.1 %	< 0.1	5724.910637	-0.81 %
1000	8067.4	< 0.1 %	< 0.1	7983.128758	-1.1 %
1100	1.1233×10^4	< 0.1 %	< 0.1	11081.084249	-1.4 %
1200	1.5576×10^4	< 0.1 %	< 0.1	15311.088000	-1.7 %
1300	2.151×10^4	< 0.1 %	< 0.1	21053.744000	-2.2 %
1400	2.960×10^4	0.1 %	< 0.1	28801.928000	-2.7 %
1500	4.058×10^4	0.1 %	< 0.1	39195.464000	-3.5 %
1600	5.550×10^4	0.2 %	< 0.1	53049.504000	-4.4 %
1700	7.569×10^4	0.3 %	< 0.1	71405.104000	-5.7 %
1800	1.031×10^5	0.4 %	0.2	95576.712000	-7.4 %
1900	1.404×10^5	0.6 %	0.4	127212.960000	-9.4 %
2000	1.91×10^5	0.9 %	0.8	168386.400000	-12 %
2100	2.60×10^5	1.2 %	1.7	221649.840000	-15 %
2200	3.54×10^5	1.7 %	3.5	290181.040000	-18 %
2300	4.82×10^5	2.3 %	7	377872.080000	-22 %
2400	6.58×10^5	3.0 %	12	489486.880000	-26 %
2500	8.97×10^5	3.8 %	20	630824.480000	-30 %
2600	1.22×10^6	4.8 %	30	808914.880000	-34 %
2700	1.67×10^6	5.7 %	50	1032217.520000	-38 %
2800	2.27×10^6	6.7 %	65	1311021.600000	-42 %
2900	3.09×10^6	7.5 %	80	1657300.800000	-46 %
3000	4.19×10^6	8.3 %	120	2085888.000000	-50 %

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